

PA4 Change Log

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Trần Huỳnh Mạnh Đạt

1. Use Case Specification AI Feature Update

Performed by: Trần Huỳnh Mạnh Đạt | **Reviewed by:** Đào Duy Anh | **Edited by:** Trần Huỳnh Mạnh Đạt

Updated file: PA4/Use Case Specification.md
Update date: 08 August 2026
Purpose: Add the implemented Dormify AI chatbot feature that was missing from the PA4 Use Case Specification.

Change ID	Section updated	Change made	Repository evidence
PA4-UCS-AI-01	Document metadata and snapshot	Updated version to 1.1 – PA4 AI update, date to 08 August 2026, scope to PA4, use-case count from 72 to 76, functional groups from 11 to 12, and added Dormify AI as the selected PA4 addition.	PA4/Use Case Specification.md
PA4-UCS-AI-02	PA4 Structure Checklist	Renamed the checklist from PA3 to PA4 and clarified that PA4 AI use cases use repository evidence because no PA3 AI prototype screenshot exists.	Domitory_Management_Frontend/app/components/ChatbotWidget.tsx
PA4-UCS-AI-03	Table of Contents and Functional Group Index	Added 13. AI Chatbot Assistance and renumbered traceability appendices to sections 14-17. Added the AI functional group with 4 use cases.	PA4/Use Case Specification.md
PA4-UCS-AI-04	Introduction and actors	Added a modeling decision that Dormify AI is active because ChatbotModule is registered in the backend root module. Added Authenticated User and Ollama AI Runtime actors, and updated	Domitory_Management_Backend/src/app.module.ts, Domitory_Management_Backend/src/chatbot/chatbot.module.ts

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		Student/System Admin responsibilities.	
PA4-UCS-AI-05	New section 13	Added the AI Chatbot Assistance functional group with four use cases: UC-AI-01 Ask Dormify AI Question , UC-AI-02 Submit AI Answer Feedback , UC-AI-03 Review AI Answer Feedback , and UC-AI-04 Rebuild AI Knowledge Base .	<code>Domitory_Management_Backend/src/chatbot/chatbot.controller.ts</code> , <code>Domitory_Management_Backend/src/chatbot/chatbot.service.ts</code>
PA4-UCS-AI-06	UC-AI-01	Defined authenticated chatbot asking, SSE streaming, recent-history handling, vector plus keyword retrieval, source chips, not-found guidance, personalized profile/contract/invoice context, and structured invoice-card display.	<code>ChatbotController.streamChat</code> , <code>ChatbotService.prepareContext</code> , <code>ChatbotService.searchKnowledgeDetailed</code> , <code>ChatbotService.getPersonalContext</code> , <code>ChatbotService.getInvoiceCard</code> , <code>ChatbotWidget.tsx</code>
PA4-UCS-AI-07	UC-AI-02	Defined thumbs-up/thumbs-down answer feedback, optimistic frontend state, backend validation, and upsert behavior keyed by user and question.	<code>ChatbotController.submitFeedback</code> , <code>ChatbotService.saveFeedback</code> , <code>ChatFeedbackSchema</code>
PA4-UCS-AI-08	UC-AI-03	Defined administrator review of chatbot feedback, negative-only filtering, admin role protection, sorting, and populated user identity fields.	<code>GET /api/chatbot/feedback, @Roles('ADMIN'), ChatbotService.listFeedback</code>
PA4-UCS-AI-09	UC-AI-04	Defined administrator knowledge-base rebuild from Markdown documents, chunking with source labels, Ollama embedding generation, MongoDB knowledge storage,	<code>POST /api/chatbot/ingest, ChatbotService.ingestData, KnowledgeSchema, scripts/run-ingest.ts</code>

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		and normalized keyword-search text.	
PA4-UCS-AI-10	Traceability appendices	Added FR31, ST29–ST30, and MT09 traceability rows for Dormify AI, then updated the document end note to Version 1.1 / PA4.	PA4/Use Case Specification.md

2. Verification

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Check	Result
Detailed use-case headings	76
Use-case rows in functional group tables	76
New AI use-case IDs	UC-AI-01 to UC-AI-04
Functional groups after update	12
Traceability sections after renumbering	Sections 14-17
Backend evidence checked	ChatbotModule, ChatbotController, ChatbotService, KnowledgeSchema, ChatFeedbackSchema, ingestion scripts